

Week 2 Practical

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```
mysql> CREATE DATABASE bookflow_db;  
ERROR 1007 (HY000): Can't create database 'bookflow_db'; database exists
```

```
mysql> USE bookflow_db;  
Database changed
```

```
mysql> CREATE TABLE Books (  
-> book_id INT PRIMARY KEY,  
-> title VARCHAR(100) NOT NULL,  
-> isbn VARCHAR(20) UNIQUE,  
-> published_year INT CHECK (published_year < 2027)  
-> );  
ERROR 1050 (42S01): Table 'books' already exists
```

```
mysql> create database bookflow;  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> CREATE TABLE Books (  
-> book_id INT PRIMARY KEY,  
-> title VARCHAR(100) NOT NULL,  
-> isbn VARCHAR(20) UNIQUE,  
-> published_year INT CHECK (published_year < 2027)  
-> );  
ERROR 1050 (42S01): Table 'books' already exists
```

```
mysql> use bookflow;  
Database changed
```

```
mysql> CREATE TABLE Books (  
-> book_id INT PRIMARY KEY,  
-> title VARCHAR(100) NOT NULL,  
-> isbn VARCHAR(20) UNIQUE,  
-> published_year INT CHECK (published_year < 2027)  
-> );  
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> INSERT INTO Books (book_id, title, isbn, published_year) VALUES  
-> (1, 'The Great Gatsby', '9780743273565', 1925),
```

-> (2, 'To Kill a Mockingbird', '9780061120084', 1960),

-> (3, '1984', '9780451524935', 1949);

Query OK, 3 rows affected (0.01 sec)

Records: 3 Duplicates: 0 Warnings: 0

mysql> select * from books;

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949

3 rows in set (0.00 sec)

```
mysql> CREATE TABLE Members (  
-> member_id INT PRIMARY KEY,  
-> full_name VARCHAR(100),  
-> email VARCHAR(100) UNIQUE  
-> );
```

Query OK, 0 rows affected (0.07 sec)

mysql> INSERT INTO Members (member_id, full_name, email) VALUES

-> (101, 'John Smith', 'john.smith@email.com'),

-> (102, 'Emma Wilson', 'emma.wilson@email.com'),

-> (103, 'Michael Brown', 'michael.brown@email.com');

Query OK, 3 rows affected (0.01 sec)

Records: 3 Duplicates: 0 Warnings: 0

mysql> select * from members;

member_id	full_name	email
101	John Smith	john.smith@email.com
102	Emma Wilson	emma.wilson@email.com
103	Michael Brown	michael.brown@email.com

3 rows in set (0.00 sec)

```
mysql> CREATE TABLE Loans (  
-> loan_id INT PRIMARY KEY,  
-> member_id INT,  
-> book_id INT,  
-> loan_date DATE,
```

```
-> FOREIGN KEY (member_id) REFERENCES Members(member_id),
-> FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );
```

Query OK, 0 rows affected (0.07 sec)

```
mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
-> (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
-> (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
-> (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
-> (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
-> (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
```

Query OK, 10 rows affected (0.01 sec)

Records: 10 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM Loans;
```

```
+-----+-----+-----+-----+
| loan_id | member_id | book_id | loan_date |
+-----+-----+-----+-----+
| 1 | 101 | 1 | 2025-01-05 |
| 2 | 102 | 2 | 2025-01-08 |
| 3 | 103 | 3 | 2025-01-10 |
| 4 | 101 | 2 | 2025-02-01 |
| 5 | 102 | 1 | 2025-02-05 |
| 6 | 103 | 2 | 2025-02-12 |
| 7 | 101 | 3 | 2025-03-01 |
| 8 | 102 | 3 | 2025-03-07 |
| 9 | 103 | 1 | 2025-03-15 |
| 10 | 101 | 1 | 2025-04-01 |
+-----+-----+-----+-----+
```

10 rows in set (0.00 sec)

```
mysql> SELECT m.full_name AS Member_Name, b.title AS Book_Title
-> FROM Loans l
-> INNER JOIN Members m ON l.member_id = m.member_id
-> INNER JOIN Books b ON l.book_id = b.book_id;
```

```
+-----+-----+
| Member_Name | Book_Title |
+-----+-----+
| John Smith | The Great Gatsby |
| John Smith | To Kill a Mockingbird |
| John Smith | 1984 |
| John Smith | The Great Gatsby |
| Emma Wilson | To Kill a Mockingbird |
| Emma Wilson | The Great Gatsby |
```

Emma Wilson	1984
Michael Brown	1984
Michael Brown	To Kill a Mockingbird
Michael Brown	The Great Gatsby

+-----+-----+

10 rows in set (0.00 sec)

```
mysql> SELECT published_year, COUNT(book_id) AS Total_Books
```

```
-> FROM Books
```

```
-> GROUP BY published_year
```

```
-> ORDER BY published_year;
```

+-----+-----+

published_year	Total_Books
----------------	-------------

+-----+-----+

1925	1
------	---

1949	1
------	---

1960	1
------	---

+-----+-----+

3 rows in set (0.00 sec)

```
mysql> CREATE TABLE Donation_History (
```

```
-> donation_id INT PRIMARY KEY,
```

```
-> book_id INT,
```

```
-> donor_name VARCHAR(100),
```

```
-> donation_date DATE,
```

```
-> FOREIGN KEY (book_id) REFERENCES Books(book_id)
```

```
-> );
```

Query OK, 0 rows affected (0.07 sec)

```
mysql> START TRANSACTION;
```

Query OK, 0 rows affected (0.00 sec)

```
mysql> INSERT INTO Books (book_id, title, isbn, published_year)
```

```
-> VALUES (4, 'Animal Farm', '9780451526342', 1945);
```

Query OK, 1 row affected (0.00 sec)

```
mysql> INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
```

```
-> VALUES (1, 4, 'Raj Kumar', CURDATE());
```

Query OK, 1 row affected (0.00 sec)

```
mysql> commit;
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> CREATE INDEX idx_books_isbn ON Books(isbn);
```

Query OK, 0 rows affected (0.12 sec)
Records: 0 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM Books WHERE isbn = '9780451524935';
```

book_id	title	isbn	published_year
3		9780451524935	1949

1 row in set (0.00 sec)